EXPLORATORY DATA ANALYSIS USING POWER BI

WEEK 10

REAL-ESTATE ANALYTICS DASHBOARD

REPORT

Submitted by

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Project Overview:

Dataset: King County House Sales Dataset (Pre-processed from Week 9)

Tool Used: Microsoft Power BI Desktop

Objective:

To build an interactive dashboard to help real-estate managers understand property prices, locations, house characteristics, and sales patterns.

Problem Statement:

Real-estate managers do not have a consolidated view of property performance. They need a single dashboard to answer: Which locations have highest prices? How do bedrooms, bathrooms, grade, condition and waterfront affect price? What is the sales trend over time?

DAX Measures :

Measures Table

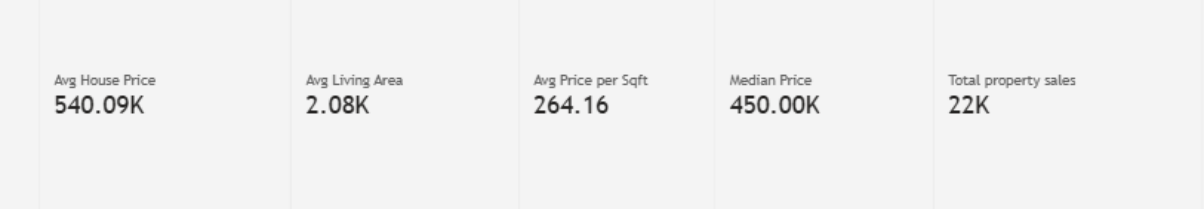
Created empty table Measures using Enter Data.

- Total property sales = COUNTROWS('kc_house_data')
- Avg House Price = AVERAGE(kc_house_data[price])
- Median Price = MEDIAN(kc_house_data[price])
- Avg Living Area = AVERAGE(kc_house_data[sqft_living])
- Avg Price per Sqft = AVERAGE(kc_house_data[Price per sqft])

Dashboard Visual Layout - House Sales Analytics Dashboard:

Cards:

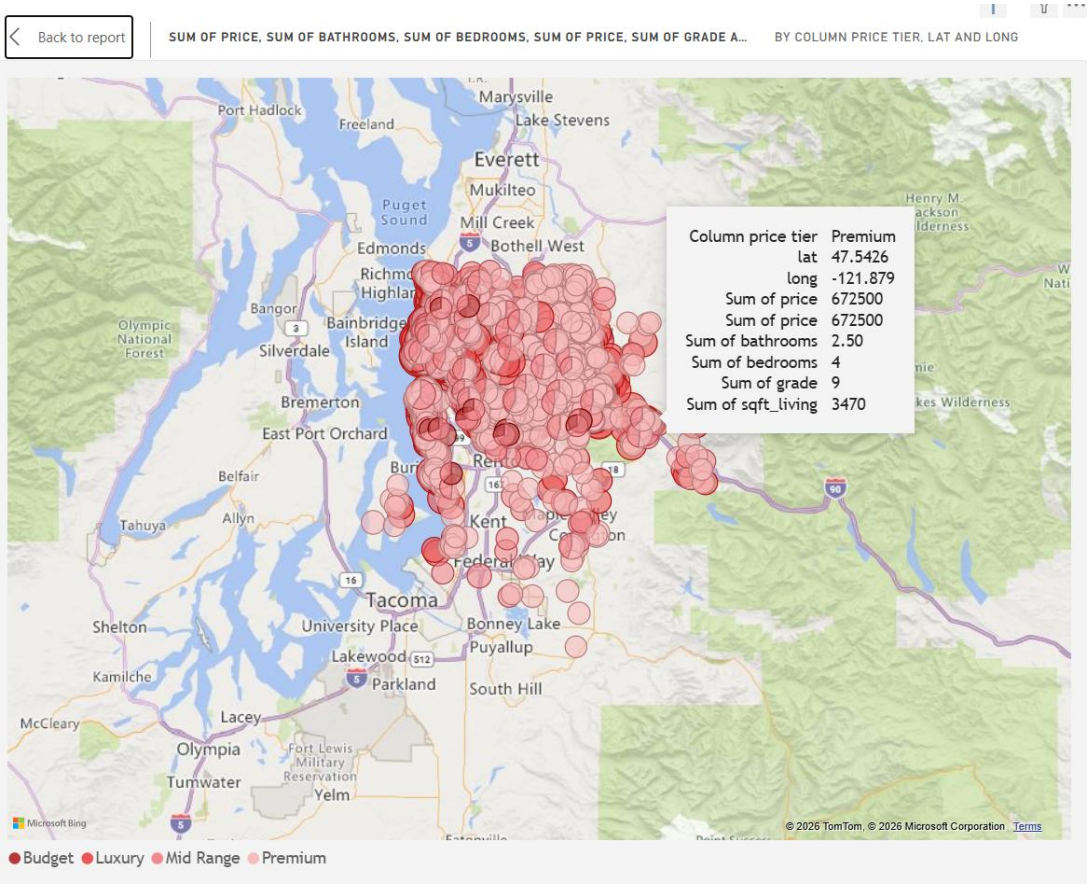
- Total Properties Sales
- Avg House Price
- Median Price
- Avg Living Area
- Avg Price per Sqft



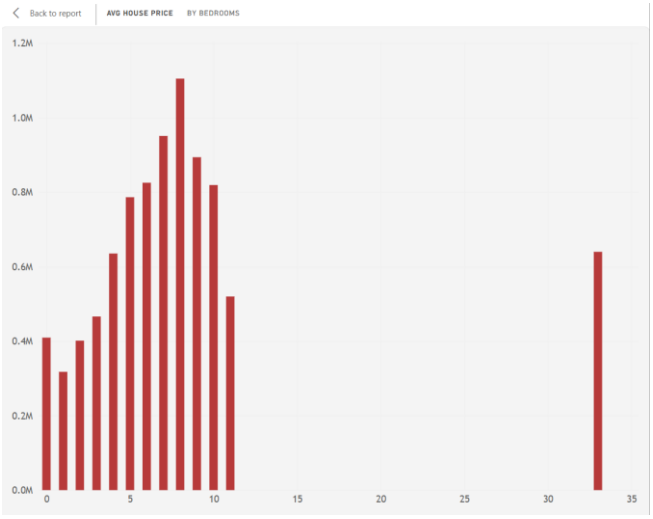
Spatial & Price Analysis:

Visual 1: Map – Latitude = lat, Longitude = long, Bubble Size = Sum of price, legend = Column price tier,

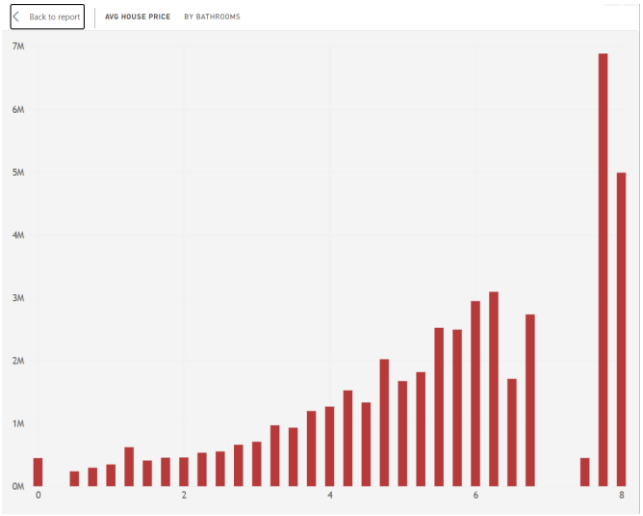
Tooltips = Sum of price,bathrooms,bedrooms,grade and Sqft_living.



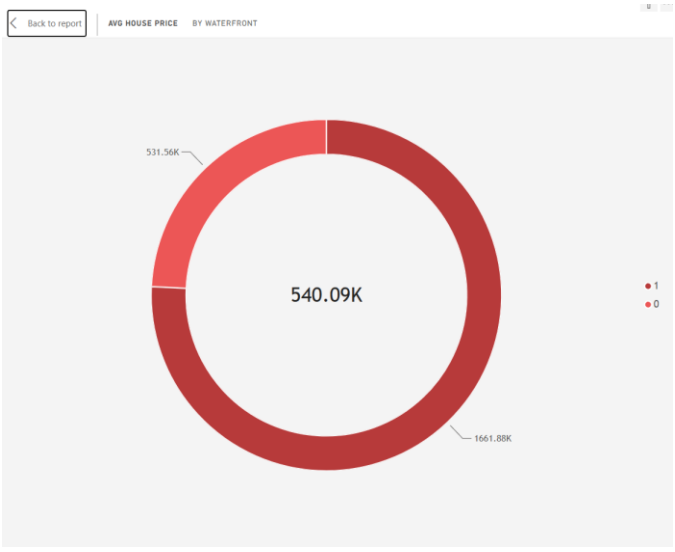
Visual 2: Clustered Column Chart - bedrooms vs Avg House Price



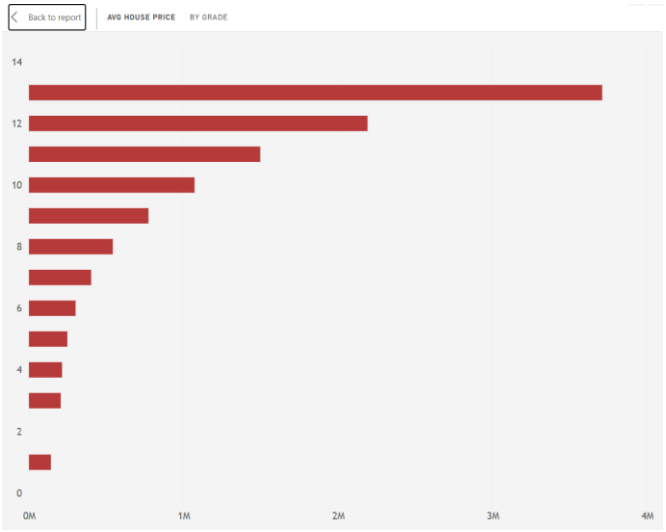
Visual 3: Clustered Column Chart - bathrooms vs Avg House Price



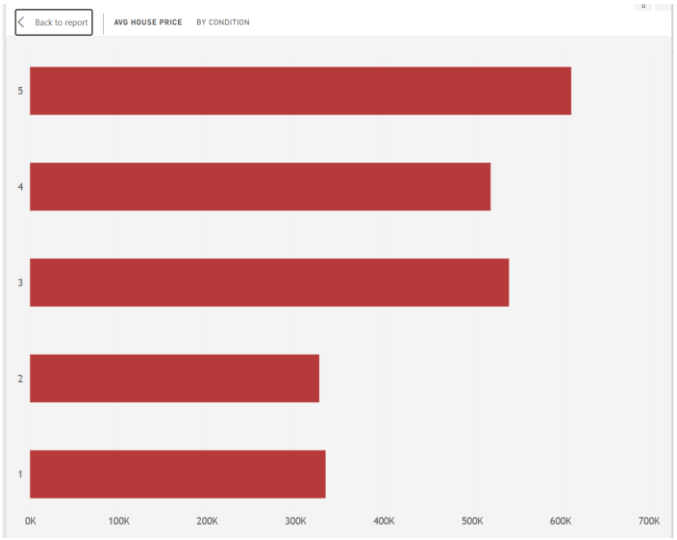
Visual 4: Donut Chart - waterfront vs Avg House Price



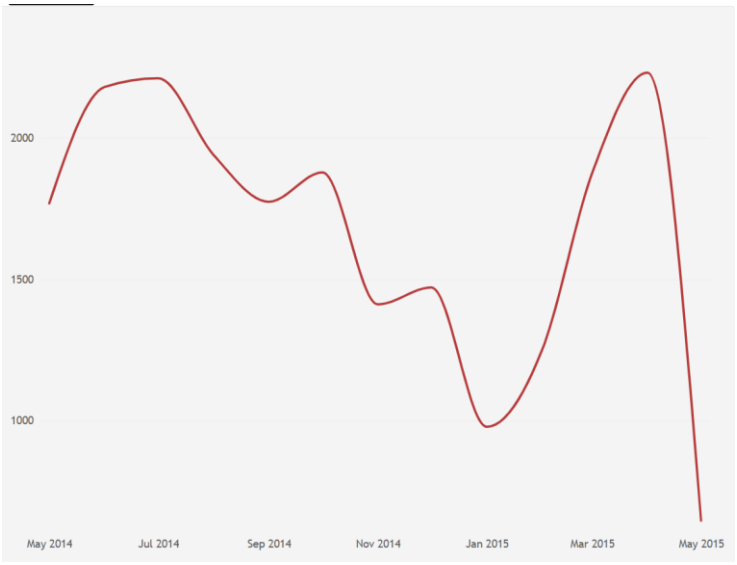
Visual 5: Stacked Bar Chart - grade vs Avg House Price



Visual 6: Stacked Bar Chart - condition vs Avg House Price

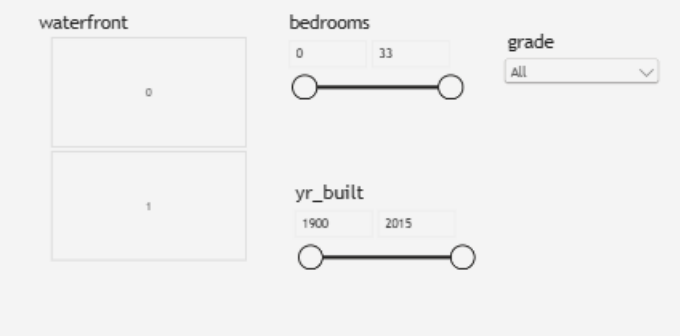


Visual 7: Line Chart - Monthly Sales Trend (Year -> Month) vs Total Properties Sold



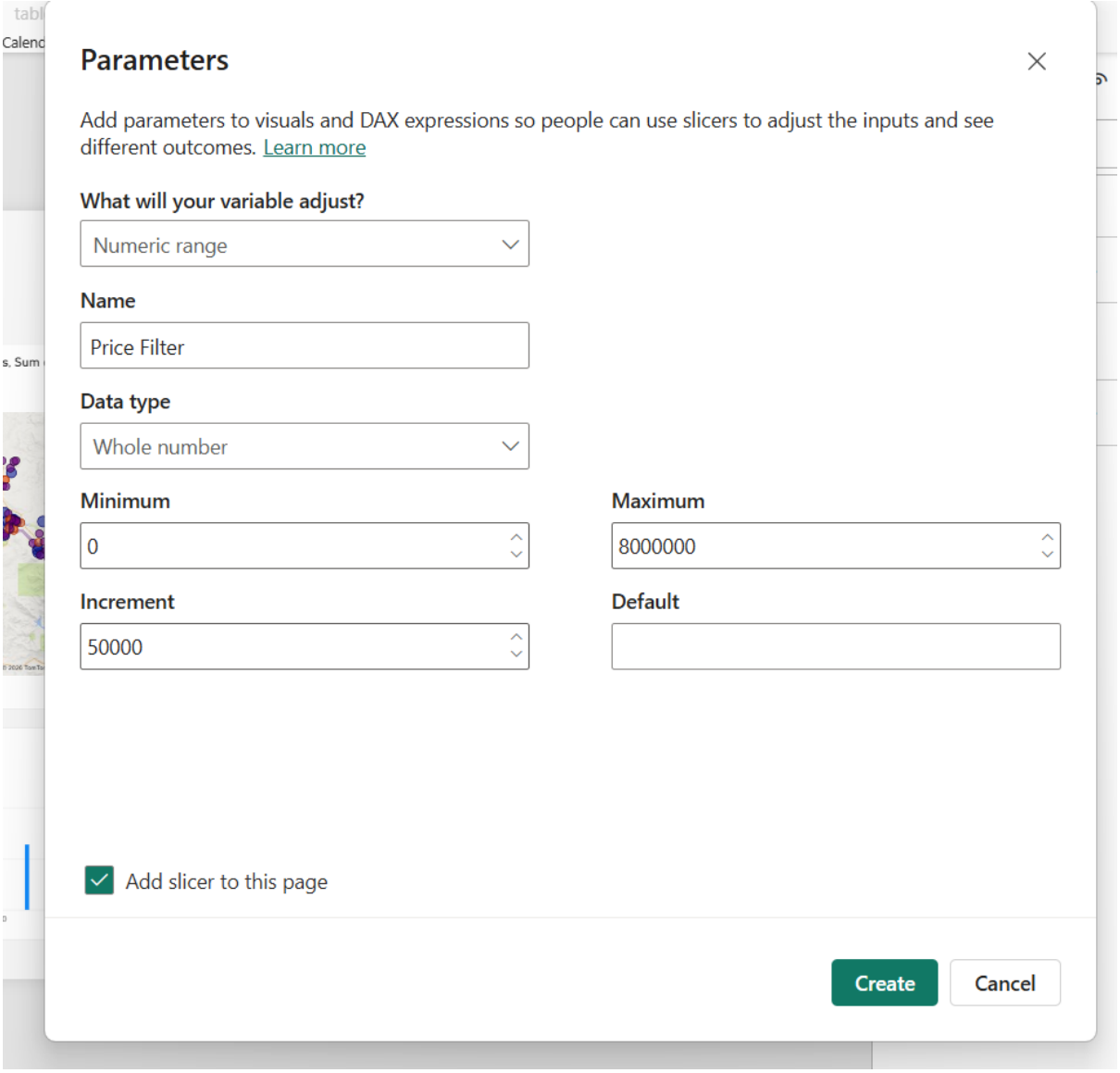
Interactivity Features:

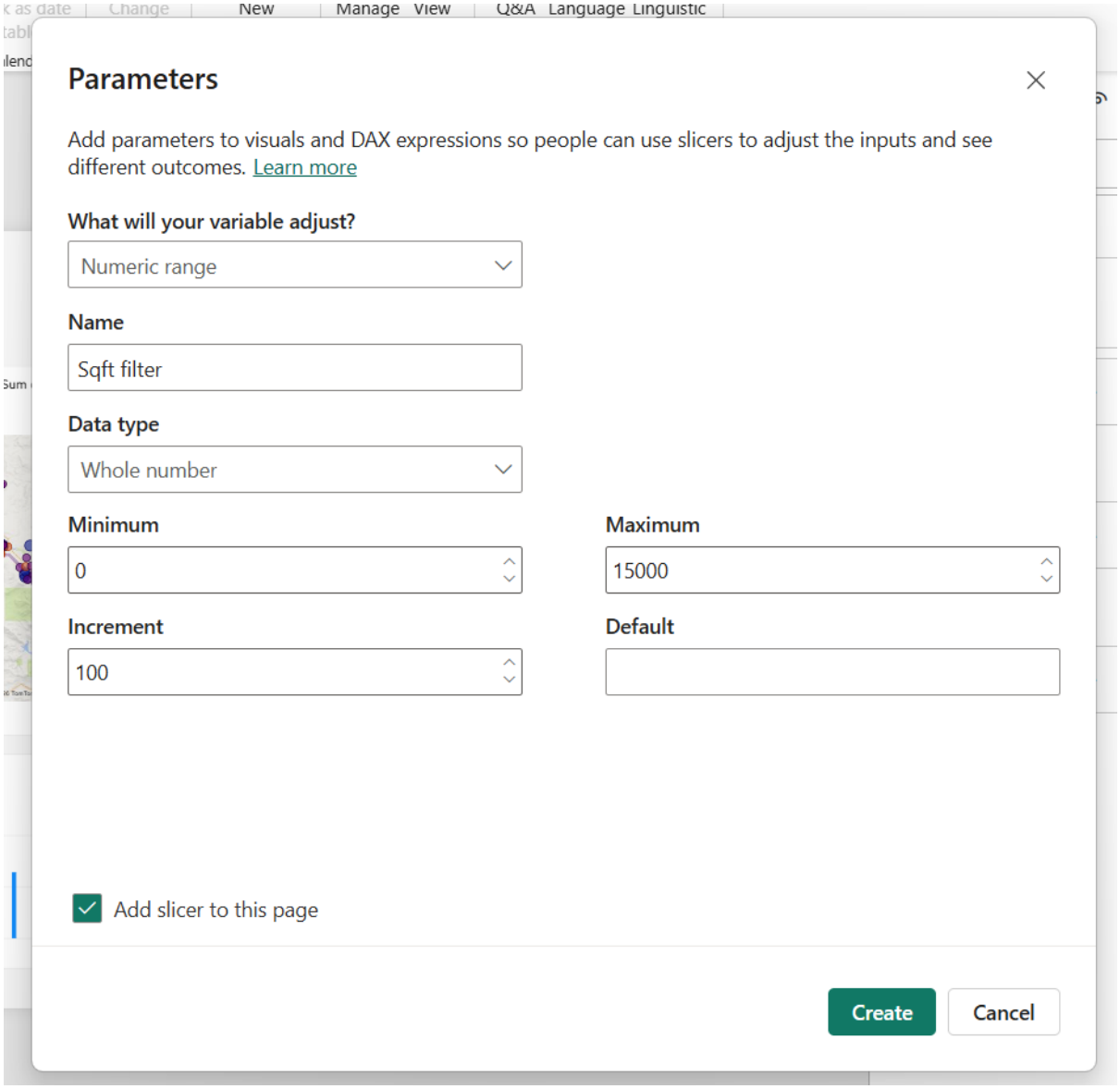
Slicers: waterfront, grade, bedrooms, yr_built



Numeric Range Parameters:

Price Filter Parameter (0 to 8000000) and Sqft Filter Parameter (0 to 15000) created in Modeling -> New Parameter.





Added as sliders.

Moving slider dynamically filters all charts.

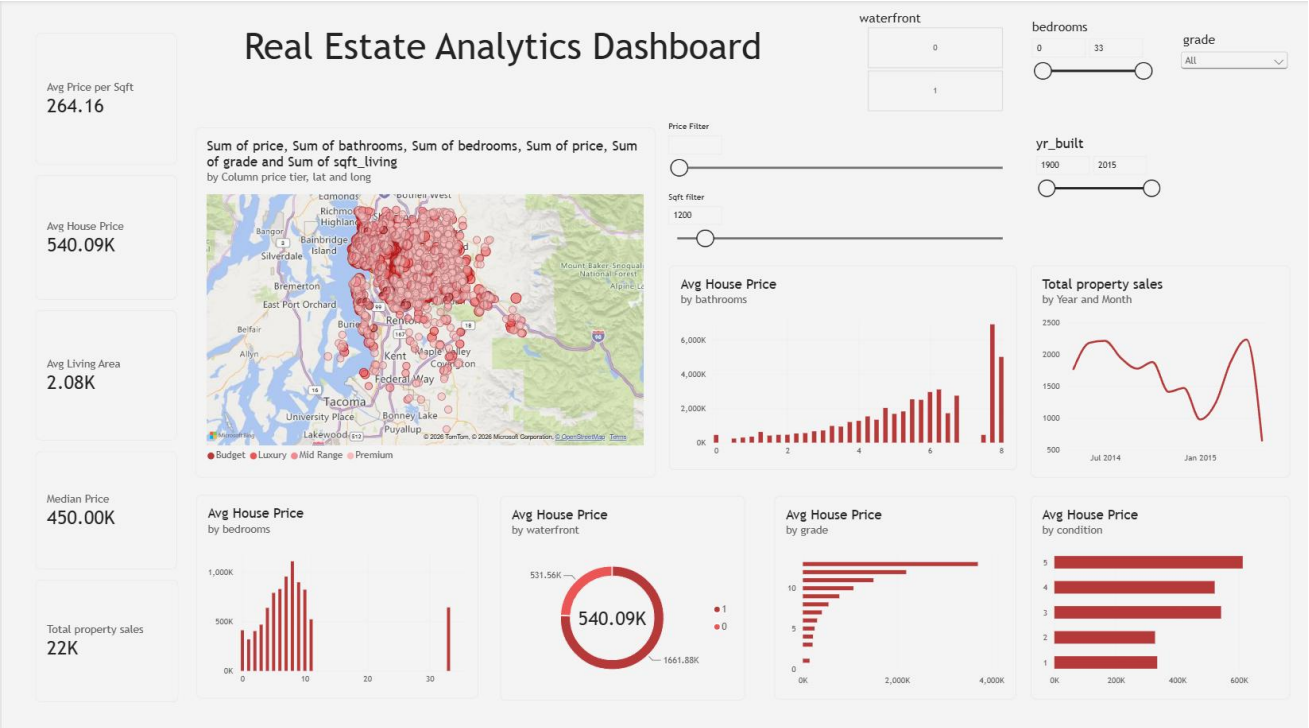
Price Filter:



Sqft filter:



Final Dashboard:



Key Insights:

- Waterfront properties have almost 2x higher average price than non-waterfront properties.
- Houses with Grade 10 and above command highest prices.
- Price increases with bedrooms up to 4, then stabilizes.
- Houses near city centre (low lat/long variation) show higher price per sqft.
- Peak sales observed in summer months.

Conclusion:

Week 10 task successfully converts cleaned house sales data into a business-ready interactive dashboard. DAX measures provide dynamic Cards, Map shows spatial price patterns, and parameter sliders enable dynamic price filtering for managers.